Exp No:11

28/9/26

Routing Network Layer.

AIM:- a) Simulate static routing configuration using CISCO packet Tracer

Procedure:-

Static routes are the routes you manually add to the router's routing table. The process of adding static routes to the routing table is known as static routing.

#ip route 30.0.0.0 255.0.0.0 20.0.0.2 10.
#ip route 30.0.0.0 255.0.0.0 40.0.0.2 20.

router 0 configuration:

router>enable
router# configure
terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router (config) #ip route 30.0.0.0 255.0.0.0 20.0.0.2 10
Router (config) #ip route 30.0.0.0 255.0.0.0 40.0.0.2 20.
Router (config) #exit.

Press the CLI prompt of router 0 and run the following commands.

router 1 configuration:-

router >enable
Router # configure
terminal
Enter configuration commands, one per line. End with CNTL/Z
router (config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1 10
router (config)#ip route 10.0.0.0 255.0.0.0 1 20.
Router (config) #exit

Router 2 configuration:

router >enable
Router #configure
terminal
Enter configuration commands, one per line, End with
CNTL/Z.
Router (config)#ip route 10.0.0.0 255.0.0.0 40.0.0.1

observed:- Static routes to 10.0.0.0/8 and 30.0.0.0/8
were added sucessfully.

Verification:-

Pc> ping 30.0.0.100
Pc> tracert 30.0.0.100

To check the routing table:
show ip route static.

To test the backup route, disconnect the router0 - router1 link and run:

Pc> ping 200.0.0.100

Pc> tracert 30.0.0.100

Simulate RIP using Cisco Packet Tracer.

observation:-

configure router()

router>enable

router# configure terminal

router(config)# interface fast ethernet 0/1

router(config-if)# ip address 10.0.0.1 255.0.0.0

Router(config-if)# no Shutdown.

Router(config-if)# exit.

Router(config)# interface serial 0/0/0

Router(config-if)# ip address.

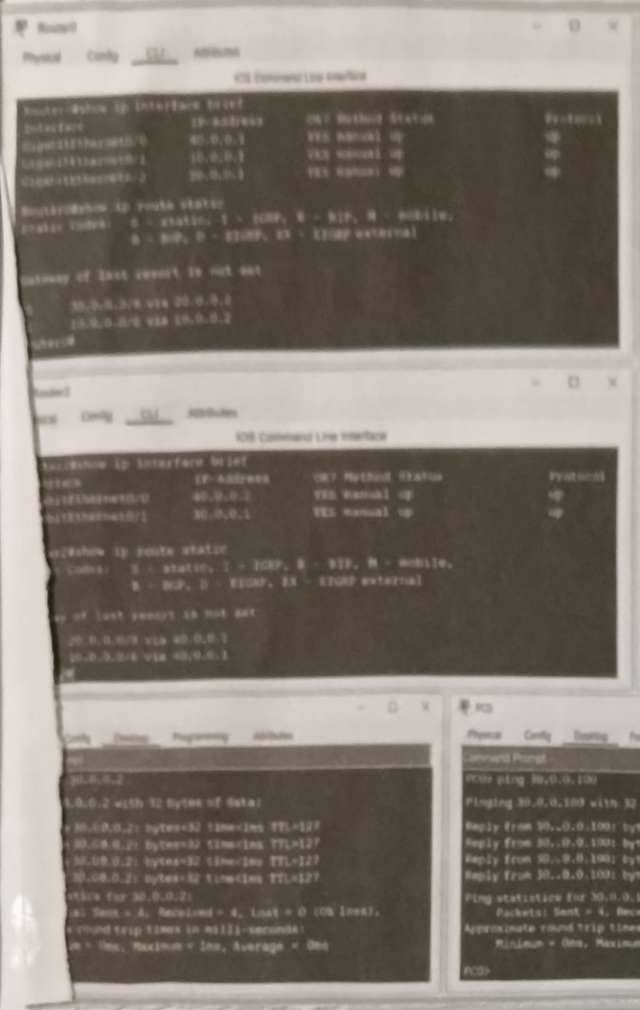
192.168.1.249 255.255.255.252

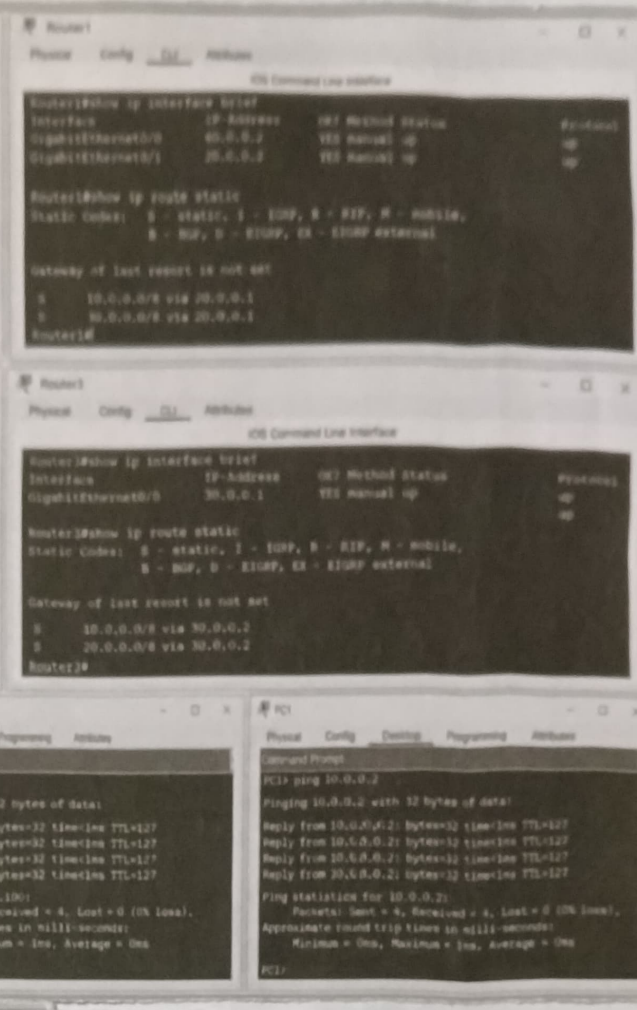
Router(config-if)# clock rate 64000

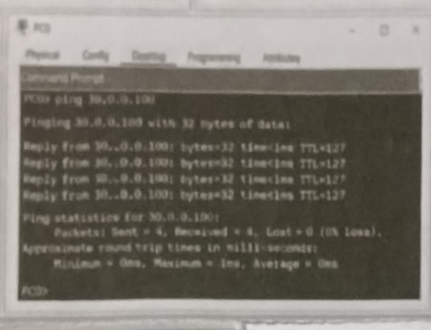
Router(config-if)# bandwidth 64

Router(config-if)# no shut down

Router(config-if)# exit.







PC0 — Command Prompt (ping)

```
C:\> ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=1ms  TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms  TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms  TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms  TTL=126

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\> tracert 20.0.0.2
Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1      1 ms      1 ms      1 ms   10.0.0.1
  2      2 ms      2 ms      2 ms   192.168.1.253
  3      3 ms      3 ms      3 ms   20.0.0.2

Trace complete.

C:\>
```

Result:

RIP was successfully configured and verified. The network automatically selected the route with the minimum count.